



Pesticide exposure risks and seasonal variation in honey bee-collected pollen across agricultural landscapes in Latvia

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Abstract

Honey bees (*Apis mellifera*) are widely used as bioindicators of environmental contamination and bee-collected pollen reflects pesticide exposure in agricultural landscapes. This study evaluated pesticide residues in honey bee-collected pollen from six Latvian apiaries during 2024 and 2025 flowering seasons, and assessed toxicological risk using the Pollen Hazard Quotient (PHQ). In total, 15 active substances were detected in 2024 and 19 in 2025. Fungicides were the most frequently detected pesticide class in both years, accounting for 51% of detections in 2024 and 54% in 2025, followed by herbicides (38% and 24%) and insecticides (11% and 22%). The highest concentrations were recorded for glyphosate, fluazifop-P-butyl, boscalid, azoxystrobin, acetamiprid, and tebuconazole. Seasonal variation was evident, with higher residue concentrations and PHQ values mainly from mid-May to early July, coinciding with crop flowering and pesticide application periods. Median PHQ values remained below the threshold (PHQ < 50), however several samples in 2025 exceeded this level, with a maximum PHQ of 1173.91 driven by cypermethrin. No consistent descriptive pattern was observed between PHQ values and the proportion of agricultural land or nectar producing crops. This suggests that landscape composition alone did not explain differences in pesticide exposure among the studied apiaries. The results indicate that pesticide exposure is influenced by temporal variation, pesticide toxicity, and local factors such as crop management and foraging behaviour, rather than landscape composition alone. Honey bee-collected pollen proved to be a suitable tool for monitoring pesticide exposure. It also allowed the identification of periods of increased risk in agricultural environments.

Keywords *Apis mellifera* · Pesticide residues · Pollinator exposure · Agricultural landscapes · Pollen hazard quotient

Introduction

Global agricultural production increasingly relies on insect pollination due to the expansion of pollinator dependent crops (Aizen et al. 2019). A similar trend can be observed

in Latvia, where the cultivated area of oilseed rape has remained consistently high in recent years (145.1 thousand ha), while the area of legume crops has expanded by 84% since 2020 (Official Statistics Portal of Latvia 2026). However, bee populations continue to decline due to multiple environmental stressors associated with agricultural intensification (Winfree 2010; Lima et al. 2022).

Pesticide exposure and limited floral resources are among the main stressors affecting managed and wild bee populations. Mass flowering crops may increase food availability for pollinators, but can also increase exposure to pesticide residues. This interaction remains insufficiently studied (Goulson et al. 2015; Mallick et al. 2023). This is particularly relevant for oilseed rape (*Brassica napus*), where foliar fertilization and pesticide application can affect flower visiting bees, including honey bees (*Apis mellifera*), bumble bees (*Bombus* spp.), and solitary bee species (Viik 2012).

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Honey bees and their products are widely recognised as effective bioindicators of environmental contamination, including heavy metals and persistent organic pollutants, such as pesticides (Bargańska et al. 2016). Bee-collected pollen is particularly informative for pesticide exposure studies because it reflects material collected by foraging bees from flowering plants. It can also be linked to both pesticide residues and botanical composition. Compared with wax or bee bread, freshly collected pollen provides a more temporally resolved indication of recent exposure. Nectar can also indicate pesticide exposure, but it does not provide the same direct link with pollen botanical composition. Recent studies have shown that pesticide residues are also present in the pollen and nectar of wildflowers. This may increase pollinator exposure to pesticides, sometimes at concentrations higher than those detected in cultivated crops (Botías et al. 2015).

Most pesticides are applied during cultivation of agricultural crops. Bees are primarily exposed to these substances near farmland, particularly during crop flowering periods (Calatayud-Vernich et al. 2019). In Latvia, the flowering period for several major agricultural crops occurs between May and June and coincides with intensive pesticide use. Previous studies have linked neonicotinoid seed treatments in oilseed rape to negative effects on wild bee populations, highlighting the importance of assessing pollinator exposure to pesticides (Woodcock et al. 2016). Despite increasing attention to pesticide exposure in pollinators, information on pesticide contamination in honey bee-collected pollen in relation to landscape composition remains limited in the Baltic region. This region is particularly relevant because the flowering period of major crops, including oilseed rape and legumes, overlaps with intensive pesticide use, while regional long term monitoring data remain scarce.

This study follows earlier research initiated and published in 2024 (Ozols et al. 2024), and extends the dataset to include the 2024 and 2025 sampling seasons. By combining seasonal pesticide residue data, PHQ-based risk screening, and landscape composition analysis, this study provides a more detailed assessment of pollinator exposure under Baltic agricultural conditions. Our main objective was to further assess pesticide contamination in honey bees collected pollen, and to evaluate the potential risks to bee health in relation to surrounding landscape composition. To achieve this objective, we focused on four main tasks: (1) identifying pesticide residues in honey bee-collected pollen samples; (2) examining the seasonal dynamics of residue occurrence; (3) screening and evaluating the potential impact of these residues on bee health; and (4) investigating the relationship between landscape structure, expressed as the proportion of agricultural land, and pesticide related risks to bees. Two hypotheses were formulated. We hypothesised that PHQ values would increase with increasing agricultural land cover

and that pesticide occurrence would peak during the main flowering period from May to June.

Materials and methods

Study area and landscape assessment

Six commercial apiaries representing different regions of Latvia were selected for this study (Fig. 1), and pollen samples were collected from these sites in seasons 2024 and 2025. Three of the selected apiaries were located in Zemgale region, which represents the most intensive agricultural production in Latvia: GPS coordinates for Jelgava town (56°39'46.4" N, 23°45'14.2" E), Platone parish (56°30'47.5" N, 23°41'03.8" E), and Vecauce parish (56°28'03.2" N, 22°53'16.4" E). The remaining three apiaries were situated in the Vidzeme, Kurzeme, and Latgale: Lube parish (57°26'23.3" N, 22°35'33.9" E), Ledurga parish (57°17'33.1" N, 24°49'07.0" E), and Ambeli parish (56°05'19.9" N, 26°53'47.4" E).

Landscape composition was assessed within a 3-km radius around each apiary. This radius was selected to represent the main foraging area of honey bees under typical agricultural conditions. Although honey bees can forage longer distances, including within six- to 8-km radius, foraging activity is generally concentrated closer to the hive when floral resources are available (Beekman and Ratnieks 2000). Spatial processing and landscape classification were performed using QGIS (version 4.0. "Norrköping"). Landscape variables were treated as continuous variables and expressed as the proportion (%) of each land use class within the 3-km radius around each apiary. Detailed information on the biotopes surrounding the study sites is provided in Online Resource 1. Agricultural land data were obtained from the Rural Support Service of Latvia, forest cover data from the State Forest Registry, and urban area estimates from the Dynamic World dataset (Brown et al. 2022). Mean values for the study years 2024 and 2025 were extracted from the dataset, and the total area classified under the "built-up area" was calculated. In this study, the category "built-up area" was referred to as "urban", additional data are given in Online Resource 2. As pesticide residues were expected to occur primarily within agricultural areas, all forested land, natural and semi-natural grasslands (originally classified under agricultural land), as well as waterbodies and similar habitats, were grouped into a category "semi-natural". Areas that did not fall within the agricultural, urban, or forest classes were also assigned to the "semi-natural" category. Consequently, three main landscape classes were defined: Semi-natural, Urban and Agriculture. Each site was classified to a dominant landscape category when a single land use class exceeded 50% of the surrounding area. Dominant



Fig. 1 Locations of pollen sampling sites and classification of land use categories (Agriculture—landscapes dominated by agricultural land; Urban—landscapes dominated by built up areas and infrastructure elements; Seminatual—landscapes dominated by forests, shrubs,

grasslands, waterbodies and similar biotopes). The numbering of sites (1–6) corresponds to the sampling locations presented in Online Resource 2

categories were used only for descriptive site classification. In Jelgava town, semi natural and urban land uses occurred in comparable proportions because of the extensive green infrastructure, including parks, public green areas and tree covered spaces. Nevertheless, the site was classified as urban due to the location of the apiary, even though “build up” land represented only 44% of the area.

Pollen collection and palynological analysis

Pollen sampling was conducted from 15 May to 20 August in 2024 and from 5 May to 10 August in 2025. Depending on weather conditions and pollen availability, collections took place every second day or at longer intervals, using pollen traps installed at the apiaries. Samples affected by rain were excluded, and damaged material was removed after each collection. The pollen traps were then cleaned with a 70% ethanol solution before the next sampling event. At each apiary, pollen was collected from two marked hives equipped with pollen traps. Two hives per apiary were used to reduce colony level variability while maintaining a consistent sampling design and limiting disturbance to the colonies. Pollen collected from two hives was pooled into one daily sample. A target sample mass of approximately 100 g was collected

when sufficient pollen was available. This target mass was used to standardise sample handling, although the actual amount depended on pollen availability during each collection. Each sample was labelled with the apiary name and date of collection. After fieldwork, the samples were placed in a cooler container with ice elements for temporary storage and transported to the laboratory. Then, they were stored frozen at -18°C until analysis.

In both study years, the sampling season was divided into seven two-week periods. In 2024, the sampling periods were as follows: 15–28 May (period 1), 29 May–11 June (period 2), 12–25 June (period 3), 26 June–9 July (period 4), 10–23 July (period 5), 24 July–6 August (period 6), and 7–20 August (period 7). In 2025, the periods were slightly adjusted: 5–18 May (period 1), 19 May–1 June (period 2), 2–15 June (period 3), 16–29 June (period 4), 30 June–13 July (period 5), 14–27 July (period 6), and 28 July–10 August (period 7). All daily samples from the same apiary and period were combined into one composite two-week sample. This resulted in one composite sample per apiary for each two-week sampling period. No additional independent field replicates were collected within the same apiary and period. In total, 41 pollen samples were analysed for 2024—seven from each apiary, except for period 1 at the Ambeli

parish site. At this location, bees began foraging only at the end of May because of limited availability of floral resources earlier in the season. In 2025, 40 pollen samples were analysed, with seven samples from each apiary except for period 4 at the Jelgava town site and period 6 at the Platone parish site. In contrast to the missing Ambeli parish sample in 2024, these missing samples in 2025 were not related to delayed seasonal foraging. These samples could not be obtained because pollen collection was insufficient during the respective sampling periods.

The botanical composition of pollen samples was analysed at the laboratory of the Institute of Plant Protection Research “Agrihorts”. A modified light microscopy approach was applied (Jones and Bryant 1998; Halbritter et al. 2018). Pollen identification and counting were carried out using images captured under a light microscope at 400× magnification. For each composite sample, three replicate palynological analyses were conducted. In each replicate, 8–10 images were evaluated, with approximately 10–40 pollen grains per image. This resulted in more than 500 identified pollen grains per sample.

Pesticide residue analysis in pollen samples

The chemical analysis of pollen samples was performed by Bilacon GmbH, a QS-recognised and DAkkS-accredited analytical laboratory (Berlin, Germany). According to the laboratory reports, pesticide residues and related compounds were analysed using validated combi methods PV-SA-085 and PV-SA-343 Teil A. Method PV-SA-085 included multi residue screening by liquid chromatography tandem mass spectrometry (LC-MS/MS), gas chromatography tandem mass spectrometry (GC-MS/MS) and GC-NCI. Method PV-SA-343 Teil A was applied for the analysis of glyphosate and AMPA by LC-MS/MS.

Each pollen sample was analysed under controlled laboratory conditions. The analytical scope for method PV-SA-085, supplied by laboratory, included the target pesticide list together with reporting limits, limits of quantification (LOQ), limits of detection (LOD), and the analytical platform used for each compound (Online resource 3). The scope document listed 793 active substances and related compounds. The reporting limit for most substances was 0.01 mg kg⁻¹.

The laboratory confirmed that the methods are accredited under ISO/IEC 17025 and are performed according to SANTE requirements. Measurement uncertainty (U) was reported by the laboratory where applicable. According to the laboratory decision rule, expanded measurement uncertainty was calculated using a coverage factor $k=2$, corresponding to a confidence level of approximately 95%. The laboratory also confirmed that recovery values were within the 70 to 130% range. Full internal validation reports,

including compound specific recovery data, matrix effects, internal standards, calibration ranges and precision data, were not provided to the authors because they were considered sensitive laboratory information.

According to the laboratory reports, GC-NCI was used for selected GC-based determinations. NCI refers to negative chemical ionisation rather than to the detector itself. The exact mass spectrometric detector configuration for GC-NCI was not further specified in the reports available to the authors.

Pollen Hazard Quotient as an indicator of pollen toxicity

The Pollen Hazard Quotient (PHQ) was applied as a screening indicator of potential pollen toxicity to honey bees (Végh et al. 2022; Carlson et al. 2022). PHQ was calculated as the ratio between the detected pesticide concentration in pollen (ppb; µg kg⁻¹) and the corresponding oral or contact LD₅₀ for honeybees (Friedle et al. 2021). LD₅₀ values for honey bees (*Apis* spp.) were obtained from the Pesticide Properties DataBase, using entries based on EU regulatory and evaluation data published by the European Commission and European Food Safety Authority (EFSA) (European Food Safety Authority 2026). Only entries classified as verified data used for regulatory purposes were used when available. Acute oral and contact LD₅₀ values for honey bees, reported as µg bee⁻¹, were applied depending on data availability. The LD₅₀ endpoint applied for each active substance is provided in Online Resource 6. The screening approach assumes a maximum daily intake of up to 9.5 mg of bee bread per nurse bee (Crailsheim et al. 1992). PHQ values were interpreted as screening categories of potential risk: values below 50 were considered low risk, values between 50 and 500 moderate risk, and values above 500 high risk (Stoner and Eitzer 2013; Böhme et al. 2018).

All cultivated crops within the surrounded 3-km landscapes were grouped according to pesticide application intensity, based on the latest national data for Latvia (Official Statistics Portal of Latvia 2025). Four pesticide use intensity categories were used: no use (no pesticides applied), low (< 0.5 kg a.s. ha⁻¹) (a.s.—active substance), moderate (0.5–1.0 kg a.s. ha⁻¹), and high (> 1.0 kg a.s. ha⁻¹) as shown in Online Resource 4. These categories were then used to examine how pesticide input levels related to both total agricultural land cover and the proportion of nectar producing crops within the study landscapes. For this analysis, nectar plants were defined as cultivated crops visited by honey bees for both nectar and pollen collection. Given that honey bees pollinate many “key” crops, for example, oilseed rape, field beans, and apple orchards (Geslin et al. 2017; Barda et al. 2024; Reilly et al. 2024; Yang et al. 2024), we therefore

examined whether the proportion of these crops within the landscape affected values of PHQ across apiary locations.

Descriptive analyses were performed in R version 4.4.0 (R Core Team, 2024). Due to the limited number of study sites and high variability in PHQ values, no linear regression model was applied to evaluate the relationship between landscape composition and PHQ values. Instead, cumulative PHQ values were summarised descriptively by apiary, sampling period and study year. Relationships between PHQ values and the proportion of agricultural land, nectar producing crops and pesticide use intensity categories within a radius of 3 km around each apiary were evaluated using graphical comparison.

Results and discussion

Active substances detected and quantified in pollen samples

In pollen samples collected during study year 2024, 15 pesticide active substances were detected and quantified. The detected residues were dominated by fungicides, which accounted for eight active substances. Among these, boscalid and azoxystrobin were recorded most often, occurring in six samples (14.6%) and four samples (9.8%), respectively. Herbicides represented the second most common group, with six active substances identified, of which glyphosate and clopyralid occurred most frequently, each being detected in six samples (14.6%). Only one insecticide, acetamiprid, was detected, occurring in five samples (12.2%). The detected substances were consistent with crop protection practices in the main agricultural crops surrounding the apiaries, including oilseed rape, legumes such as field beans and peas, cereals and orchard crops. In 2025, pollen samples contained residues of 19 active substances. In comparison to 2024, herbicides formed the largest group, again comprising eight active substances, with glyphosate detected most frequently, occurring in six samples (15.0%). This was followed by seven fungicides, among which azoxystrobin, occurring in 11 samples (27.5%), while boscalid and tebuconazole were each detected in nine samples (22.5%), respectively. In addition, four insecticides were detected, with acetamiprid showing the highest occurrence among this group, being detected in 10 samples (25.0%) (see Table 1).

Previous investigators have consistently reported fungicides as dominant residues in pollen. For example, a study in Slovakia and Germany by Sabo et al. (Sabo et al. 2024) reported maximum pollen concentrations of 40.1 $\mu\text{g kg}^{-1}$ for boscalid and 520.61 $\mu\text{g kg}^{-1}$ for azoxystrobin. Meanwhile a Polish monitoring study documented even higher concentrations, reaching 150 $\mu\text{g kg}^{-1}$ for boscalid and 2900 $\mu\text{g kg}^{-1}$ for azoxystrobin (Swiatly-Blaszkiewicz

et al. 2025). This pattern was consistent with our results, where fungicides were the most diverse pesticide group in 2024 and remained among the most frequently detected substances in 2025. Boscalid and azoxystrobin reached maximum concentrations of 610 and 480 $\mu\text{g kg}^{-1}$ in 2024 and 360 and 76 $\mu\text{g kg}^{-1}$ in 2025, respectively.

Tebuconazole was also frequently detected in 2025 and reached 240 $\mu\text{g kg}^{-1}$, indicating continued fungicide input during the flowering season. The higher tebuconazole level observed in 2025 may be related to the timing of triazole fungicide applications in surrounding crop fields, particularly oilseed rape, cereals, or orchard crops. Residue occurrence in pollen may also have been influenced by bee foraging on flowering crops or adjacent flowering vegetation during the same period. Meanwhile, in Sweden, Nicholson et al. (Nicholson et al. 2024) reported infrequent detections of tebuconazole, with concentrations reaching 227 $\mu\text{g kg}^{-1}$. Higher concentrations have also been reported from Romania, where tebuconazole levels in bee pollen reached 120 $\mu\text{g kg}^{-1}$ (Bodescu et al. 2025).

Herbicides were also an important group, particularly glyphosate and clopyralid in 2024. Glyphosate reached 840 $\mu\text{g kg}^{-1}$ in 2024 and 327 $\mu\text{g kg}^{-1}$ in 2025, which was higher than values reported in several European pollen studies (Kaila et al. 2022; Jesús et al. 2024). These elevated values may reflect differences in sampling timing, local herbicide use practices, and the analytical approach used for highly polar compounds. The highest glyphosate concentrations occurred during periods when bees were actively collecting pollen in agricultural landscapes. This suggests that local pesticide application timing may have contributed to observed levels. In Latvia, high glyphosate concentrations have also been reported in bee related matrices, including honey and bee bread (Lace et al. 2026). However, notably higher substance levels have been reported in other bee related matrices from Latvia, where concentrations in honey and bee bread reached 1400 $\mu\text{g kg}^{-1}$ and 1900 $\mu\text{g kg}^{-1}$, respectively. Our findings indicate that glyphosate contamination of bee products may be relevant under local agricultural conditions.

Fluazifop-P-butyl, an herbicide, was detected once in pollen samples in 2024 and three times in 2025. In both study years, this compound reached the highest concentration among all detected substances, measuring 660 $\mu\text{g kg}^{-1}$ in 2024 and 1800 $\mu\text{g kg}^{-1}$ in 2025. These levels exceeded the EU recommended MRL of 50 $\mu\text{g kg}^{-1}$ for bee products. Available literature indicates a wide range of reported fluazifop-P-butyl concentrations in bee pollen. For example, substantially higher levels were detected in pollen samples from southern Germany, where Böhme et al. (Böhme et al. 2018) reported concentrations reaching 6831.3 $\mu\text{g kg}^{-1}$. Fluazifop-P-butyl concentrations detected in this study were higher than most values reported in

Table 1 Pesticide active substances detected in honey bee-collected pollen in 2024 and 2025

Active substance	Pesticide class*	Max detected concentration, $\mu\text{g kg}^{-1}$	Min detected concentration, $\mu\text{g kg}^{-1}$	Detection frequency (> LOD)
Study year 2024				
Acetamiprid	I	350	14	5
Aclonifen	H	39	39	1
Azoxystrobin	F	480	10	4
Boscalid	F	610	24	6
Clopyralid	H	96	33	6
Difenoconazole	F	11	11	1
Dimoxystrobin	F	74	15	3
Fluazifop-P-butyl	H	660	660	1
Fluroksypyr	H	34	34	2
Glyphosate	H	840	24	6
MCPA	H	17	11	2
Mefentrifluconazole	F	66	19	3
Pyraclostrobin	F	160	83	2
Prothioconazole-desthio	F	110	15	3
Tebuconazole	F	82	12	2
Study year 2025				
2,4-D	H	17	17	1
Acetamiprid	I	200	24	10
Azoxystrobin	F	76	11	11
Boscalid	F	360	11	9
Captan	F	88	22	3
Clethodim	H	150	150	1
Clopyralid	H	80	80	1
Cypermethrin	I	27	27	1
Cyprodinil	F	87	87	1
Difenoconazole	F	19	11	4
Fluazifop-P-butyl	H	1800	20	3
Flupyradifurone	I	130	63	3
Glyphosate	H	327	12	6
Mefentrifluconazole	F	140	29	3
Metazachlor	H	28	16	2
Pyraclostrobin	F	21	10	3
Tau-fluvalinate	I	180	33	2
Tebuconazole	F	240	10	9
Terbutryn	H	19	19	1

*F fungicide, I insecticide, H herbicide

Europe, but remained within the broad range published in previous studies.

Insecticide acetamiprid was detected more frequently in 2025, occurring in 10 samples and reaching a maximum concentration of $200 \mu\text{g kg}^{-1}$. The relatively frequent occurrence of acetamiprid may reflect its use against aphids and other plant sucking pests in several crop systems surrounding the apiaries. Its systemic properties may also facilitate residue transfer into pollen when applications occur close to flowering. In 2024, detections were less frequent, however, the maximum concentration was

higher, reaching $350 \mu\text{g kg}^{-1}$. Lower maximum concentrations of acetamiprid in bee pollen have been reported elsewhere in Europe, including $34.45 \mu\text{g kg}^{-1}$ reported by Carrera et. al. (Carrera et al. 2024). In contrast, higher concentrations have been reported outside EU, for example, a study conducted in Kenya by Koech et. al. (Koech et al. 2023) reported $1202.5 \mu\text{g kg}^{-1}$. At the EU level, the recommended MRL for acetamiprid in bee products was $50 \mu\text{g kg}^{-1}$ in 2024. Following regulatory changes, the acceptable concentration was increased to $300 \mu\text{g kg}^{-1}$ in 2025. In our case, the MRL's applicable at the time

of sampling were exceeded in individual samples in both study years.

The concentrations we found are consistent with ranges reported in previous investigations, although substantial variation among regions is evident. Such variation likely reflects differences in sampling design, timing, and analytical methods; therefore, comparisons between studies should be interpreted cautiously.

One pollen sample collected in 2025 contained herbicide terbutryn, a substance banned for use in EU. This compound was detected only once and exclusively at the Jelgava town apiary. As terbutryn was not detected in earlier sampling years, its occurrence cannot be explained by current agricultural pesticide use. Possible sources of exposure include its application as a biocidal additive in outdoor construction materials or its use for aquatic algae control. Terbutryn has been reported to leach from biocide treated façade coatings and paints into runoff and wastewater systems (Bollmann et al. 2016). The presence of a wastewater treatment plant in Jelgava town within the foraging radius of the apiary may represent a potential pathway for this contamination.

Seasonal variation and landscape influence on pollen contamination

In both study years, the number of pesticide active substances detected above the limit of detection varied notably (between 0 and 7 detections) among sites and sampling periods (Fig. 2). In 2024, pesticide detections were generally concentrated in the early sampling periods, with several sites showing high numbers of detected substances during the first half of the season. We observed the highest diversity in Vecauce parish, where multiple active substances were detected in more than one sampling period, including the highest single period counts. In comparison, apiaries located in Ambeli parish showed low detection frequencies, with pesticide residues absent or detected only occasionally during the sampling season.

A different seasonal pattern was observed in 2025 in both timing and intensity. Pesticide detections were more evenly distributed across sampling periods, and several sites showed increased numbers of detected substances later in the season. Again, Vecauce parish showed the highest overall frequency of detections, with multiple periods exceeding those observed in other sites. Platone and Lube parishes also displayed increased detection frequencies during mid and late sampling periods, indicating prolonged pesticide presence in pollen. In comparison, Jelgava town and Ambeli parish maintained relatively low numbers of detected substances across most periods.

The results indicate that pesticide occurrence in pollen was more temporally extended in 2025 than 2024. Sites surrounded by intensively managed agricultural landscapes

consistently showed higher frequencies of detected active substances, whereas locations with a greater proportion of semi-natural or urban land cover exhibited lower and more irregular detection patterns.

The highest total residue concentration was recorded during the second (20.05.–02.06.) sampling period in 2024. This peak was caused mainly by samples from Vecauce parish (Fig. 3). During this period, pollen composition was dominated by Brassicaceae family, indicating intensive foraging on flowering crucifer crops, including winter oilseed rape, which represents a major pollen source for honey bees early in the season (Ouvrard and Jacquemart 2018; Poonam et al. 2022), additional data are given in Online Resource 5. Elevated residue concentrations were also recorded in the third sampling period (03.06.–16.06.), mainly in samples from Platone and Vecauce parishes. At this time, Fabaceae pollen was predominant, suggesting foraging on leguminous crops such as field beans and peas, which are known to provide high quality pollen resources (Paulino et al. 2022; Chau and Rehan 2024). A further increase in total residue concentration occurred during the sixth sampling period (15.07.–28.07.), largely influenced by samples from Platone parish. In this period, pollen originated mainly from Chenopodiaceae flowering weeds. These plants are recognised as an important pollen source for honey bee colonies in agricultural landscapes during late summer (Engel and Maricle 2021).

In 2025, elevated residue levels were recorded during the second (19.05.–01.06.) and third (02.06.–15.06.) sampling periods. These values were mainly driven by samples from Lube and Ledurga parishes. During this time, pollen composition was dominated by crucifers and rose family plants, for example, apple orchards, plums, and pears. Increased residue levels were also observed during the sixth (14.07.–27.07.) and seventh (28.07.–10.08.) sampling periods. This increase was associated with samples from Vecauce parish. Pollen collected during these periods originated mainly from Fabaceae and Brassicaceae species, including cultivated crucifers such as summer oilseed rape, which covered 136 ha within the apiary foraging area.

In both study years, crop area within the 3-km radius did not consistently correspond to the botanical composition of collected pollen (Table 2). In 2024, winter oilseed rape and legume crops were present around several apiaries (Platone parish 238 ha, Lube parish 195 ha), and Brassicaceae and Fabaceae pollen were abundant during some sampling periods. However, higher proportions of these pollen types did not always correspond to the highest pesticide residue concentrations. For example, high Brassicaceae pollen was recorded in Ledurga and Lube parishes during the second sampling period, whereas Vecauce parish showed high residue concentrations despite a lower proportion of Brassicaceae pollen. In Jelgava, a high proportion of Fabaceae

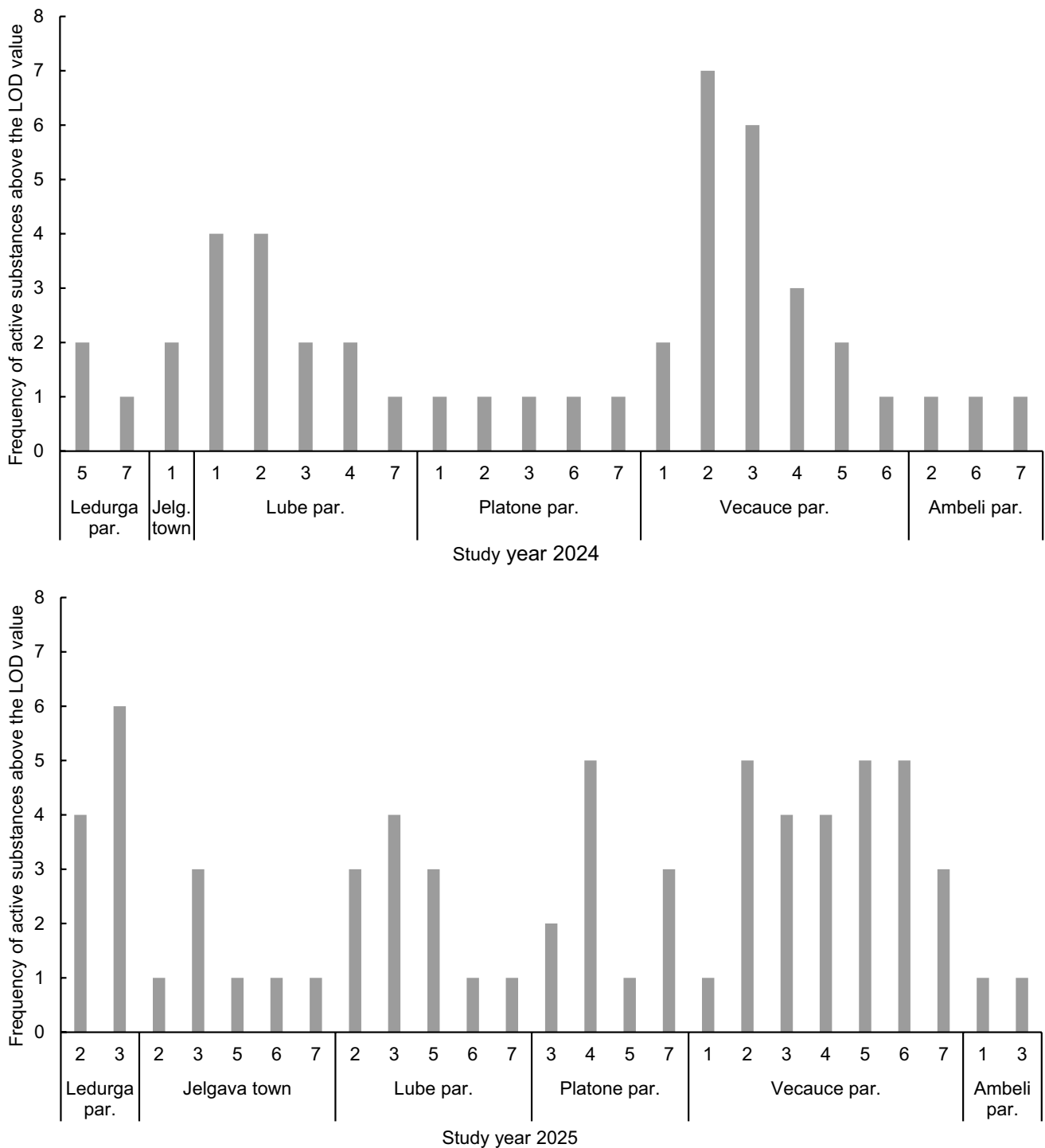


Fig. 2 Frequency of pesticide active substances detected above the LOD in honey bee-collected pollen across sampling periods and study sites in 2024 and 2025. Absence of a period for a site indicates no detections above the LOD

pollen was also recorded at apiary site, despite the absence of registered leguminous crops in the surrounding area. Therefore, Fabaceae pollen may have originated from plant species other than field beans and peas, or from more distant foraging sources.

A similar pattern was observed in 2025. Winter oil-seed rape was present around several apiaries (Platone parish 1041 ha, Ledurga 210 ha, Lube 209 ha, and Vecauce 136 ha), but the relative abundance of Brassicaceae pollen did not directly reflect the surrounding

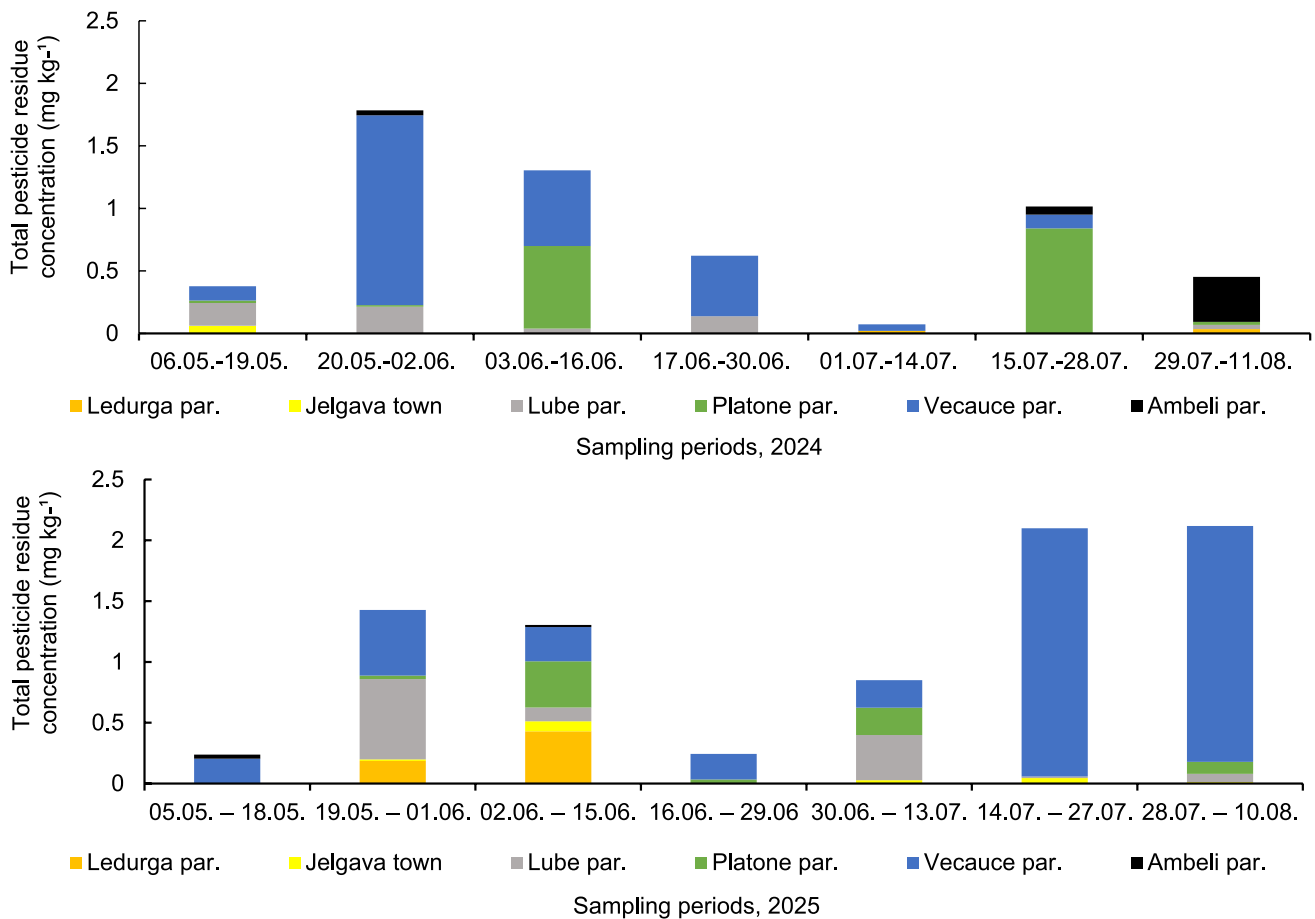


Fig. 3 Total pesticide residue concentration in honey bee pollen per sampling period and study site in 2024 and 2025

Table 2 Content of Brassicaceae, Fabaceae, and Rosaceae pollen in samples vs area of winter oilseed rape, field beans and peas, and rose family crops around the apiaries in 2024 and 2025

Study year 2024	Ledurga par	Jelgava town	Lube par	Platone par	Vecauce par	Ambeli par
Winter oilseed rape in 3-km radius, ha	35	0	195	238	121	23
Brassicaceae pollen content (%) in the pollen samples (2nd/3rd period*)	47/17	4/5	72/2	24/16	52/9	25/2
Field beans and peas in 3-km radius, ha	5	0	54	80	13	31
pollen content (%) in the Fabaceae pollen samples (3rd period*)	35	42	7	67	29	14
Study year 2025	Ledurga par	Jelgava town	Lube par	Platone par	Vecauce par	Ambeli par
Winter oilseed rape in 3-km radius, ha	210	0	209	1041	136	9
Brassicaceae pollen content (%) in the pollen samples (2nd/3rd period**)	28/78	8/40	57/58	22/58	22/34	32/55
Rose family crops in 3-km radius, ha	1	0	1	18	12	3
Rosaceae pollen content (%) in the pollen samples (2nd period**)	27	22	14	26	41	31

*2nd period to 20 May...02 June; 3rd period to 03...16 June, 2024

**2nd period to 19 May...01 June; 3rd period to 02...15 June, 2025

crop area. Rosaceae pollen was also detected at all sites, including Jelgava town, where registered orchard area was absent. This suggests that Rosaceae pollen may have

originated from urban gardens, flowering fruit trees, ornamental plants, non-crop vegetation, or more distant foraging sources.

Descriptive comparison showed that pesticide residue occurrence varied more clearly among sampling periods than among landscape categories. Higher residue concentrations were mainly observed from late May to mid-June in 2024 and during early and late season periods in 2025. These peaks coincided with periods when major flowering crops or other pollen sources were available to honey bees.

The comparison with crop area and pollen botanical composition did not show a consistent pattern across all apiaries. High proportions of Brassicaceae, Fabaceae, or Rosaceae pollen did not always correspond to the highest pesticide residue concentrations. Similarly, apiaries surrounded by larger areas of agricultural land did not consistently show higher residue levels.

Temporal factors, including crop flowering period and pesticide application timing, were more important for explaining pesticide occurrence in pollen than landscape composition alone. The systemic properties of some active substances may also facilitate residue transfer into pollen, which can weaken the direct link between mapped crop area and measured residues. Bee foraging plasticity may further contribute to this pattern, because colonies can shift between crop, non-crop, urban, and more distant floral resources depending on availability.

Pollen hazard quotient as an indicator of honey bee exposure

Cumulative PHQ values were calculated for each sampling period to estimate potential pesticide exposure of honey bees, as shown in Online Resource 6. Overall, exposure

risk was generally low, but it was characterised by short-term peaks in specific apiaries and sampling periods. In both study years, median PHQ values remained below the biologically relevant threshold of 50 in all sampling periods. However, several individual samples in 2025 exceeded this threshold, and one sample exceeded the high risk screening level of 500 (Fig. 4).

In 2024, PHQ values were generally low throughout the vegetation season. Median PHQ values ranged from 0.15 to 2.70, and no sampling period exceeded the screening threshold of 50. The highest median value (2.70) was recorded during the sixth period (15.07.–28.07). The highest cumulative PHQ value was 42.16 and was recorded in Vecauce parish during the fourth sampling period (17.06.–30.06.). This indicated a localised exposure event rather than consistent regional trend. Higher values were also observed in Vecauce parish during the second and third sampling periods, but all 2024 samples remained within the low risk category.

The PHQ results showed that toxicity, expressed by the LD_{50} values, had a stronger influence on risk screening than concentration alone. Compounds with low LD_{50} values contributed disproportionately to cumulative PHQ, even when detected at moderate concentrations. For example, acetamiprid was the main contributor to cumulative PHQ in 2024 because of its relatively low contact LD_{50} . In contrast, frequently detected fungicides and herbicides generally contributed less to PHQ because they had higher LD_{50} values.

In 2025, PHQ values remained low in most samples, but several short-term exposure peaks were observed. Ledurga parish exceeded the threshold of 50 during the second and third sampling periods, mainly due to flupyradifurone.

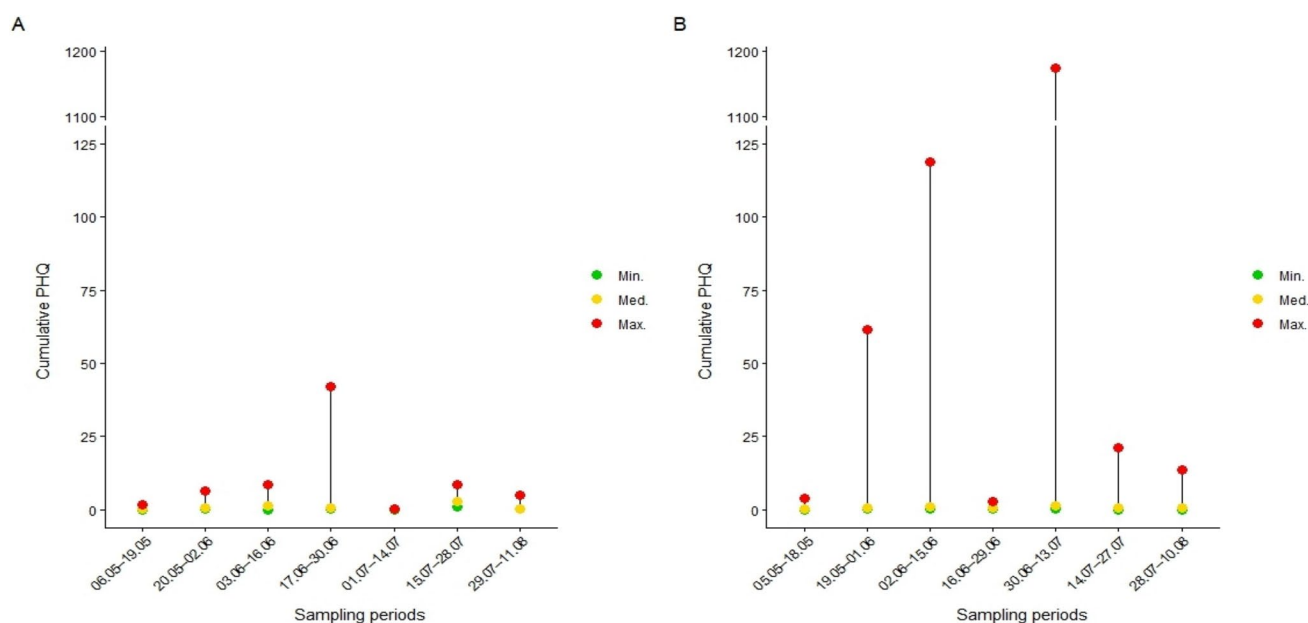


Fig. 4 Changes in cumulative PHQ in the studied honey bee apiaries in 2024 (A) and 2025 (B)

Jelgava town also exceeded this threshold during the third sampling period, driven by terbutryn. The highest cumulative PHQ value 1173.91 was recorded in Vecauce parish during the fifth sampling period and was driven almost entirely by cypermethrin. This example illustrates the importance of pesticide toxicity in PHQ interpretation: cypermethrin was detected at a moderate concentration, but its very low contact LD₅₀ resulted in a high PHQ value. Despite these peaks, median PHQ values during the same periods remained low. This indicates that elevated exposure was episodic and site specific rather than widespread across all apiaries.

PHQ is useful as a screening indicator, but it does not fully capture all biologically relevant risks. The calculation is based on acute LD₅₀ values and does not account for chronic exposure, sublethal effects or interactions among pesticide mixtures. Mixture effects were not included in the PHQ calculation. Therefore, PHQ values may underestimate real exposure risk, especially when bees are exposed repeatedly or to multiple compounds at the same time. Potential interactions among pesticide groups also cannot be excluded.

Overall, toxicological pressure at the colony level was low in 2024 and mostly low in 2025. However, several episodic exposure peaks occurred between mid-May and early July. These short-term peaks may be biologically relevant because they coincided with active brood rearing and intensive foraging periods.

Effect of landscape composition on PHQ values

We hypothesised that apiaries located in landscapes with a higher proportion of agricultural land would have higher cumulative PHQ values. However, graphical comparison

did not show a consistent pattern across study sites or years (Fig. 5). Apiaries with higher agricultural land cover did not consistently have higher cumulative PHQ values than apiaries with lower agricultural land cover. Therefore, the hypothesis that higher PHQ values would increase with increasing agricultural land cover was not supported by the descriptive comparison.

In some cases, higher PHQ values were observed in more intensively cultivated agricultural landscapes, particularly in Vecauce parish. However, this should be interpreted only as a weak tendency, because the pattern was not consistent across all apiaries. For example, Platone parish had the highest proportion of agricultural land, but did not show the highest cumulative PHQ values. In contrast, some sites with lower agricultural land cover still showed measurable or elevated PHQ values (Fig. 5).

The absence of a consistent relationship may reflect the strong influence of temporal and local factors; pesticide exposure in honey bee-collected pollen can vary throughout the flowering season depending on pesticide application timing, crop phenology and the availability of attractive flowering resources (Böhme et al. 2018; Friedle et al. 2021; Kast et al. 2024). This is consistent with the present results, where PHQ peaks were episodic and site specific rather than consistently associated with the overall proportion of agricultural land.

The analysis of nectar producing crops and pesticide use intensity also indicated that crop presence alone did not predict exposure (Figs. 6 and 7). Apiaries surrounded by larger areas of nectar producing crop or crops classified under higher pesticide use intensity did not consistently show higher PHQ values. Conversely, measurable PHQ values

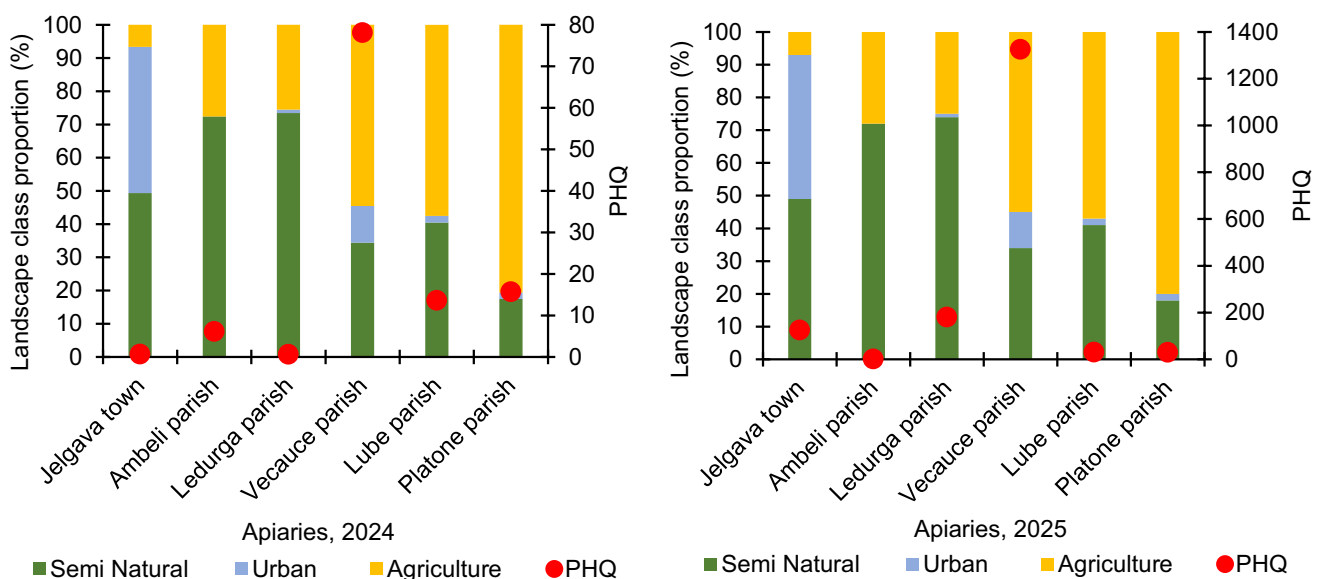


Fig. 5 Cumulative PHQ values and surrounding landscape composition within a 3-km radius of each apiary in 2024 and 2025

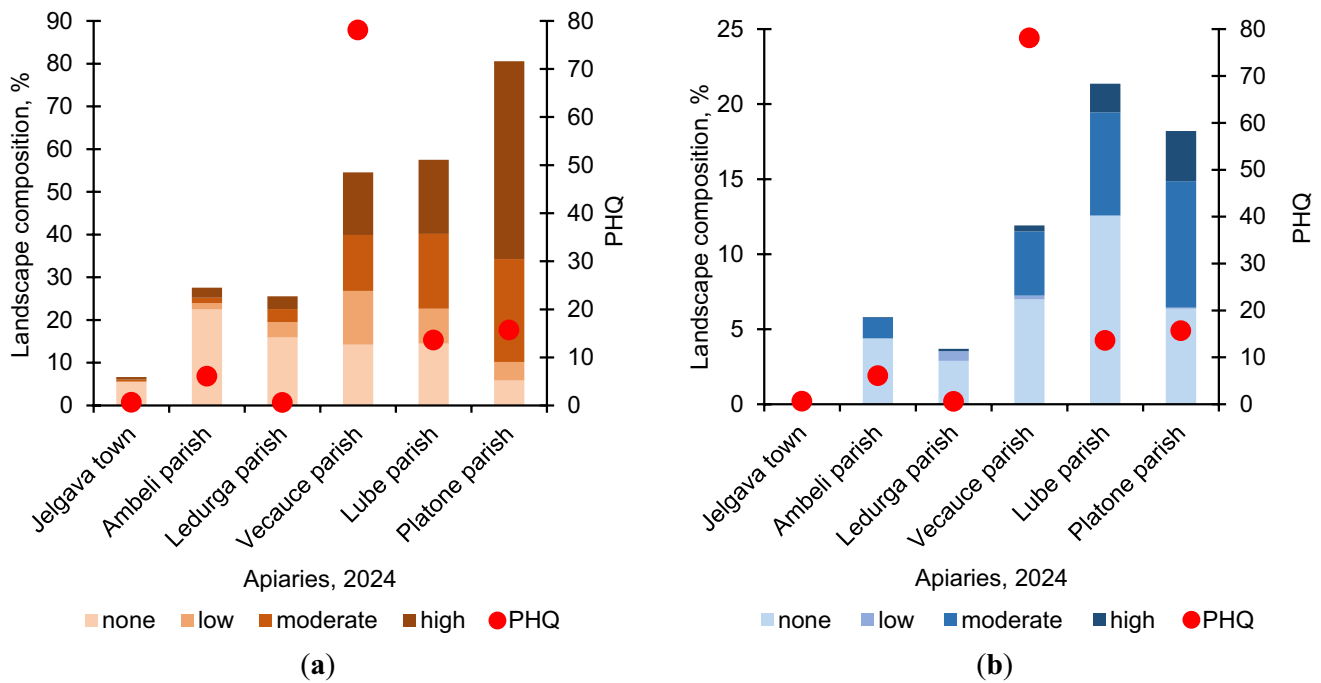


Fig. 6 Cumulative PHQ and landscape composition in a 3-km radius around the apiaries in 2024. **a** Total agricultural land; **b** nectar producing crop land. In both panels, land proportions are categorised according to pesticide use intensity

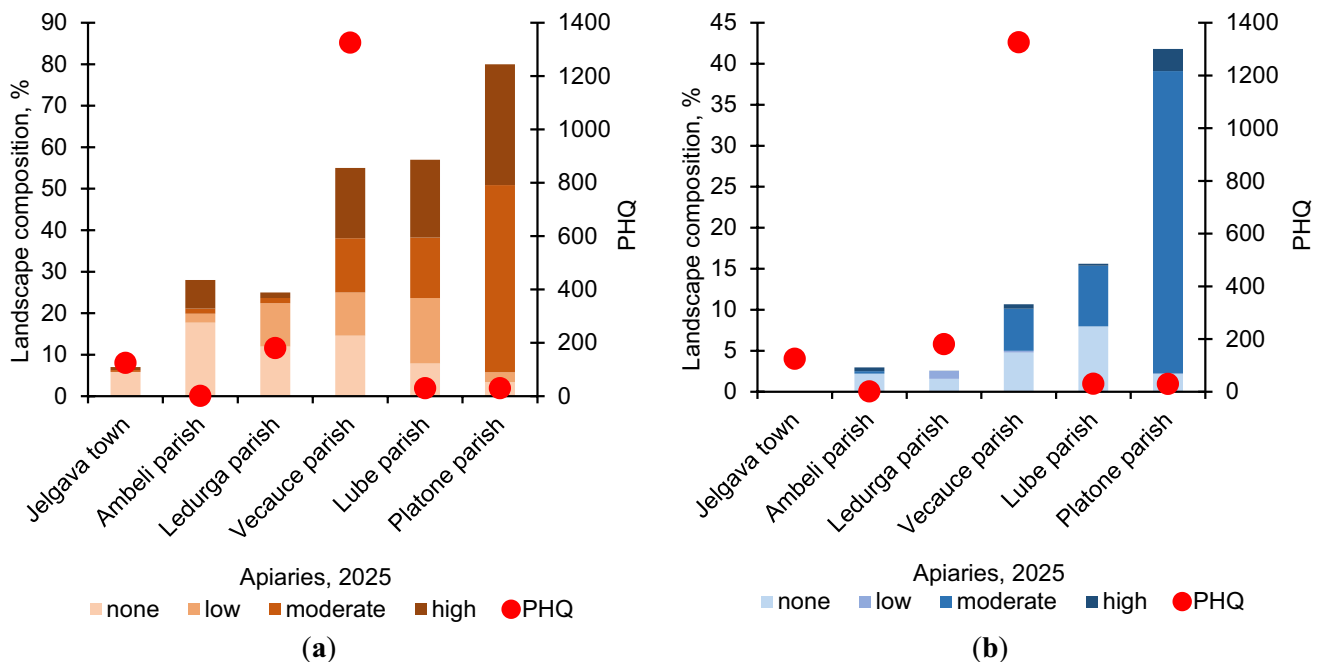


Fig. 7 Cumulative PHQ and landscape composition in a 3-km radius around the apiaries in 2025. **a** Total agricultural land; **b** nectar producing crop land. In both panels, land proportions are categorised according to pesticide use intensity

were recorded at sites with little or no presence of these categories. These observations suggest that crop distribution provides context for identifying potential exposure sources, but does not fully explain the variation in PHQ values.

Specific crop types showed a similar pattern (Fig. 8). The presence of oilseed rape, field beans, orchards or berry crops did not consistently correspond to higher cumulative PHQ values. This indicates that the presence of specific nectar

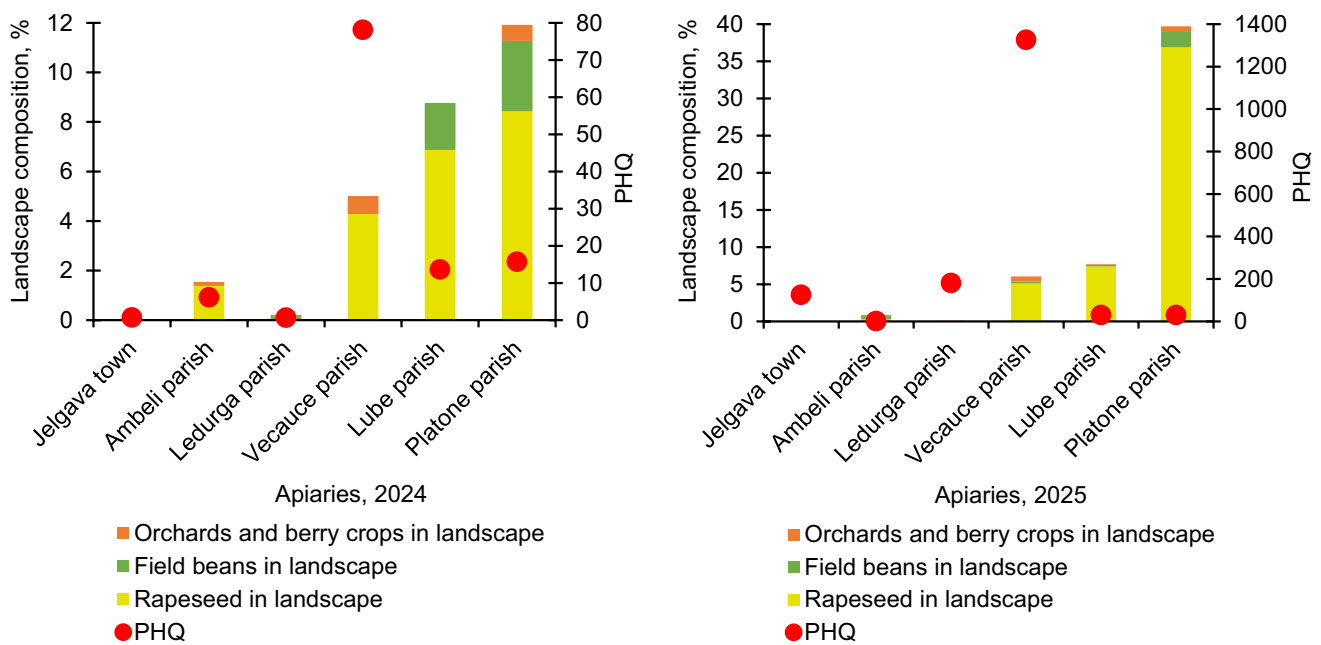


Fig. 8 Specific crop type area distribution by pesticide use intensity within a 3-km radius around the apiaries and cumulative PHQ values in 2024 and 2025

producing crops alone is insufficient to predict pesticide exposure in honey bee-collected pollen. Therefore, it is consistent with previous studies showing that pesticide residues are not explained by crop presence alone (Calatayud-Vernich et al. 2018; Schaad et al. 2023).

Several factors may explain the absence of a consistent relationship between landscape composition and PHQ values. Pesticide exposure depends on the timing of the local pesticide applications in relation to crop flowering and bee foraging activity. Honey bees may selectively forage on particular flowering resources rather than using all available crops equally, and their foraging choices may shift depending on local availability and attractiveness (Sabo et al. 2024). In addition, broad landscape categories may be too coarse to capture field level pesticide use, crop phenology and short-term exposure events.

Landscape composition provided useful context for interpreting pesticide exposure, but it did not explain PHQ patterns on its own. The observed variation was more likely driven by a combination of local pesticide application timing, crop flowering, pesticide toxicity and honey bee foraging behaviour. This interpretation is consistent with previous studies showing that surrounding land use or crop composition alone does not fully explain pesticide exposure in honey bee-collected pollen, because contamination also depends on local pesticide application practices and foraging conditions (Koech et al. 2023; Nicholson et al. 2024). Therefore, the apparent association between intensive agricultural landscapes and higher PHQ values should be interpreted

as a possible tendency rather than a statistically supported conclusion.

Conclusions

This study showed that honey bee-collected pollen contained multiple pesticide residues across agricultural landscapes in Latvia, but the overall exposure risk was generally low. Fungicides were the most frequently detected pesticide group, whereas herbicides produced some of the highest residue concentrations. However, PHQ values were driven mainly by pesticide toxicity rather than concentration alone, with highly toxic insecticides contributing disproportionately to risk screening.

In both study years, median PHQ values remained below the biologically relevant threshold of 50. Nevertheless, several short-term exposure peaks were recorded in 2025, including one high-risk PHQ value driven by cypermethrin. These results indicate that pesticide exposure in pollen was not continuous or uniform, but episodic and site specific.

Graphical comparison did not show a consistent descriptive pattern between PHQ values and the proportion of agricultural land, nectar producing crops or specific crop types around the apiaries. Landscape composition and crop distribution provided useful context for identifying potential exposure sources, but they did not explain PHQ variation on their own. Local pesticide application timing, crop flowering period, active substance toxicity

and honey bee foraging behaviour likely had stronger influence on observed exposure patterns.

This study has several limitations. The number of apiaries was limited, PHQ values were based on acute toxicity endpoints only, and no direct colony-level health measurements were included. Therefore, the results should be interpreted as a screening level assessment of potential pesticide exposure rather than a full ecological risk assessment. Site-specific meteorological variables were not included in the present analysis and should be considered in future monitoring studies.

The findings highlight the need for pesticide monitoring approaches that account not only for landscape composition but also for the timing of pesticide applications and short-term exposure peaks during crop flowering and intensive bee foraging periods.

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Author contribution All authors contributed to the study conception and design. Material preparation, data collection and analysis, methodology and investigation were performed by Niks Ozols, Maksims Filjipovičs, Betija Rubene, Sabīne Močāne, Baiba Tikuma, Valters Brusbārdis and Viktorija Zagorska. Project administration and funding acquisition was performed by Jānis Gailis and Viktorija Zagorska. Supervision and resources were performed by Niks Ozols and Viktorija Zagorska. The first draft of the manuscript was written by Niks Ozols and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The data supporting the findings of this study are included within the article, its tables, figures, and Online Resources. Additional data are available from the corresponding author upon reasonable request.

Declarations

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

Research involving human participants and/or animals This study did not involve human participants. Honey bee-collected pollen samples were obtained from managed colonies under standard beekeeping practices, and no experimental manipulation or harm to animals was performed.

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